A Hardy space approximation supporting zero-free half-planes for the ζ-function

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Abstract

Bagchi's Hilbert-space reformulation of the Báez-Duarte criterion [2] for the Riemann Hypothesis (RH) [5] states that RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of $\{G_k : k \geq 2\}$, where $G_k(s) = (k^{-s} - k^{-1})\zeta(s)/s$, in the Hardy space $H^2(\mathbb{C}_{1/2})$; here $\mathbb{C}_\alpha$ denotes the half-plane $\mathrm{Re}(s) > \alpha$. We prove that $\sum_{k=2}^{\infty} \mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$; together with the fact that membership of E in the span forces ζ to be zero-free in $\mathbb{C}_\alpha$, this gives an approximation-theoretic route recovering the classical non-vanishing of ζ on $\mathrm{Re}(s) > 1$. A matching quantitative result shows that $\alpha_c := \inf\{\alpha \geq 1/2 : d_n(\alpha)^2 \to 0\}$, the threshold at which the best n-term approximation of E converges, equals $1/2$ if and only if RH holds. Motivated by this equivalence, we numerically study the Möbius error $s_n(\alpha) = \| \sum_{k=2}^{n} \mu(k)G_k - E\|_\alpha^2$ for $1/2 \leq \alpha \leq 1$, where $\|\cdot\|_\alpha$ denotes the norm of $H^2(\mathbb{C}_\alpha)$, and the optimal error $d_n(1/2)^2$ on the critical line. An exact formula expressing $s_n(\alpha)$ as a series with elementary terms, free of the zeta function, allows us to compute it up to $n = 10^6$: on the critical line $s_n(1/2)$ oscillates without decaying, consistent with its known divergence, while for every $\alpha > 1/2$ the oscillations decay at a rate close to $n^{1-2\alpha}$.

Keywords: Riemann hypothesis, zero-free half-planes, Báez-Duarte criterion, Hardy spaces, numerical approximation.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$. Its implications for the distribution of prime numbers make it one of the central open problems in mathematics.

A functional-analytic approach to RH was initiated by Nyman and Beurling, who showed that RH is equivalent to the density of a specific family of functions in $L^2(0, 1)$ [6, 11]. Báez-Duarte

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later refined this by replacing the original uncountable family with a countable sequence, simplifying the problem while preserving its essential structure [2]. Balazard and Saias [4] extended this approach to a weighted ℓ^2-sequence space by posing the question of whether the orthogonal projection coefficients $c_{n,k}$ of $\mathbf{1}$ onto $\operatorname{span}\{r_k : 2 \le k \le n\}$ satisfy

$$\lim_{n\to\infty} c_{n,k} = -\frac{\mu(k)}{k}, \quad \forall\, k \ge 2,$$

where μ is the Möbius function and $r_k(j) = j \bmod k$ denotes the remainder upon division of j by k. This condition was shown by Weingartner in [16] to be equivalent to the RH, which identifies $-\mu(k)/k$ as the natural, and in a precise sense optimal, choice of coefficients.

A reformulation within the Hardy space $H^2(\mathbb{C}_{1/2})$ was established by Bagchi [5], where $\mathbb{C}_\alpha = \{s \in \mathbb{C} : \operatorname{Re}(s) > \alpha\}$ for $\alpha \in \mathbb{R}$: RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of

$$G_k(s) = \left(k^{-s} - k^{-1}\right)\frac{\zeta(s)}{s}, \quad k \ge 2,$$

in $H^2(\mathbb{C}_{1/2})$. Building on these ideas, Noor introduced an equivalent reformulation in the Hardy–Hilbert space $H^2(\mathbb{D})$ of the unit disk [10]: RH holds if and only if the constant function $\mathbf{1}$ belongs to the closed linear span of

$$h_k(z) = \frac{1}{1-z}\log\left(\frac{1 + z + \cdots + z^{k-1}}{k}\right), \quad k \ge 2.$$

In the same work, Noor established that the series $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I - S)h_k$ converges to $1 - z$ in $H^2(\mathbb{D})$, where S is the shift operator. For an overview of the various equivalent reformulations of the Nyman-Beurling and Báez-Duarte approaches that appear in the literature, see [7, p. 5-6].

The main idea of this paper is to generalize this result in an attempt to approximate RH using the reformulation of Bagchi in $H^2(\mathbb{C}_{1/2})$. To achieve this, we employ the Hilbert space of Dirichlet series $\mathcal{H}^2$, which is naturally connected to the Hardy space of the disk. Within this framework, we construct a bounded operator $\mathcal{M}^\tau$ that establishes a connection between $\mathcal{H}^2$ and the Hardy spaces $H^2(\mathbb{C}_\alpha)$, leading to our main result (Theorem 3.1): the series $\sum_{k=2}^{\infty} \mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$. Furthermore, Proposition 3.1 establishes that for any $\alpha \ge 1/2$,

$$E \in \overline{\operatorname{span}}\{G_k : k \ge 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \ne 0, \quad \forall\, s \in \mathbb{C}_\alpha.$$

A quantitative counterpart also holds. For $n \ge 2$ and $\alpha > 0$, let

$$d_n(\alpha)^2 := \inf_{c_2,\ldots,c_n \in \mathbb{C}} \left\| E - \sum_{k=2}^{n} c_k G_k \right\|_\alpha^2$$

denote the optimal approximation error over all coefficients. Theorem 3.2 shows that a single zero of ζ in $\mathbb{C}_\alpha$ forces $d_n(\alpha)^2$ to stay bounded away from 0, uniformly in n. Combined with Proposition 3.1, this yields Corollary 3.2: writing Θ for the supremum of the real parts of the zeros of ζ and $\alpha_c = \inf\{\alpha \ge 1/2 : d_n(\alpha)^2 \to 0\}$, one has $1/2 \le \Theta \le \alpha_c \le 1$, with $\alpha_c = 1/2$ if and only if RH holds. This recasts RH as a statement about the asymptotic behaviour of $d_n(\alpha)^2$, and is what motivates the numerical study of $d_n(1/2)^2$ in Subsection 4.3.

An important question that arises is whether the convergence of the series extends to $1/2 \leq \alpha \leq 1$. Báez-Duarte [3] showed that the $L^2(0,1)$-analog of the sequence $s_n(\alpha)$ defined below diverges, which implies the divergence of $s_n(1/2)$ itself in $H^2(\mathbb{C}_{1/2})$. Studying the convergence in the range $1/2 < \alpha \leq 1$ is therefore essential, as it could provide information about zero-free regions of ζ within the critical strip, a notoriously hard problem in its own right.

The numerical analysis conducted in this work addresses two questions. The first is whether the convergence of the partial sums

$$s_n(\alpha) = \left\| \sum_{k=2}^{n} \mu(k) G_k - E \right\|_\alpha^2$$

persists beyond the range $\alpha > 1$ settled by Theorem 3.1. To study it we express $s_n(\alpha)$ exactly as a series with elementary terms (Proposition 4.1), which avoids both the zeta function and numerical quadrature and allows us to compute $s_n(\alpha)$ up to $n = 10^6$ for $1/2 \leq \alpha \leq 1$. On the critical line $s_n(1/2)$ oscillates without decaying, consistent with the divergence result of Báez-Duarte, whereas for every $\alpha > 1/2$ the oscillations decay, at a rate close to $n^{1-2\alpha}$. The second question, guided by Corollary 3.2, is whether $d_n(1/2)^2$ itself tends to 0: since $\alpha_c = 1/2$ is equivalent to RH, tracking $d_n(1/2)^2$ against $s_n(1/2)$ tests whether the failure at $\alpha = 1/2$ reflects a genuine obstruction in the subspaces $\operatorname{span}\{G_k : k \geq 2\}$, or only in the fixed choice of coefficients $\mu(k)$. The evidence points to the latter: $d_n(1/2)^2$ decreases steadily even as $s_n(1/2)$ oscillates. Section 2 introduces the necessary function spaces and key results. Section 3 develops the main theoretical framework. Section 4 presents the numerical experiments and their implications.

2 Preliminaries

We collect the function spaces and key results needed throughout the paper. The three Hilbert spaces $H^2(\mathbb{D})$, $\mathcal{H}^2$, and $H^2(\mathbb{C}_{1/2})$ are introduced here as the analytic framework within which the completeness conditions equivalent to the Riemann Hypothesis are formulated and studied.

2.1 Hardy–Hilbert spaces and Dirichlet series

Let $H^2(\mathbb{D})$ be the Hardy–Hilbert space of the unit disk, defined as

$$H^2(\mathbb{D}) = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \|f\|_{H^2}^2 = \sum_{n=0}^{\infty} |a_n|^2 < \infty \right\}.$$

A recent reformulation of the RH, introduced in [10], is stated as follows. For each $k \geq 2$, define

$$h_k(z) = \frac{1}{1-z} \log\left(\frac{1 + z + \cdots + z^{k-1}}{k} \right).$$

Theorem 2.1 (Theorem 6, [10])**.** *The Riemann Hypothesis holds if and only if the constant function* $\mathbf{1}$ *belongs to the closed linear span of* $\{h_k : k \geq 2\}$ *in* $H^2(\mathbb{D})$.

Let $S = M_z$ denote the shift operator on $H^2(\mathbb{D})$, corresponding to multiplication by z. An additional result from [10] plays a central role in this paper.

Lemma 2.1 (Lemma 11, [10]). *The series* $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I-S)h_k$ *converges to* $1-z$ *in* $H^2(\mathbb{D})$, *where* μ *is the Möbius function.*

Let $\mathcal{H}^2$ denote the Hilbert space of Dirichlet series [8], defined as

$$\mathcal{H}^2 = \left\{ f(s) = \sum_{n=1}^{\infty} a_n n^{-s} : \|f\|_{\mathcal{H}^2}^2 = \sum_{n=1}^{\infty} |a_n|^2 < \infty \right\}.$$

Setting $H_0^2(\mathbb{D}) = H^2(\mathbb{D}) \ominus \mathbb{C}$, there exists an isometric isomorphism [8]

$$\psi : H_0^2(\mathbb{D}) \longrightarrow \mathcal{H}^2, \qquad z^k \longmapsto k^{-s}.$$

2.2 The Hardy spaces $H^2(\mathbb{C}_\alpha)$

For $\alpha \in \mathbb{R}$, let $\mathbb{C}_\alpha = \{s = \sigma + it : \sigma > \alpha\}$ and let $H^2(\mathbb{C}_\alpha)$ denote the Hardy space of analytic functions F on $\mathbb{C}_\alpha$ satisfying

$$\|F\|_{H^2(\mathbb{C}_\alpha)}^2 := \sup_{x>\alpha} \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(x+it)|^2 \, dt < \infty.$$

Any such F admits a non-tangential boundary value F^* almost everywhere on $\{\mathrm{Re}(s) = \alpha\}$, with

$$\|F\|_{H^2(\mathbb{C}_\alpha)}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F^*(\alpha+it)|^2 \, dt.$$

Moreover, $H^2(\mathbb{C}_\alpha)$ is a Hilbert space with reproducing kernel

$$k_w(s) = \frac{1}{s + \overline{w} - 2\alpha}, \quad w \in \mathbb{C}_\alpha;$$

in particular, convergence in norm implies pointwise convergence in $\mathbb{C}_\alpha$. For brevity we write $\|\cdot\|_\alpha := \|\cdot\|_{H^2(\mathbb{C}_\alpha)}$.

Define the functions G_k and E in $H^2(\mathbb{C}_{1/2})$, for $k \geq 2$, as

$$G_k(s) = (k^{-s} - k^{-1})\frac{\zeta(s)}{s} \quad \text{and} \quad E(s) = \frac{1}{s}.$$

The Riemann Hypothesis reformulation for $H^2(\mathbb{C}_{1/2})$ can be stated as follows.

Theorem 2.2 (Theorem 2, [5]). *RH is true if and only if* E *belongs to the closed linear span of* $\{G_k : k \geq 2\}$ *in* $H^2(\mathbb{C}_{1/2})$.

3 An $H^2(\mathbb{C}_\alpha)$ approximation approach

The main goal of this section is to show that E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$. For $\tau > 1/2$ and $k \geq 2$, define

$$g_{k,\tau}(z) = k^{\tau-1} \sum_{n=1}^{\infty} \frac{z^n}{n^\tau} - \sum_{n=1}^{\infty} \frac{z^{nk}}{n^\tau}.$$

It is clear that $g_{k,\tau} \in H_0^2$. Furthermore, we observe that

$$\begin{aligned}(\psi g_{k,\tau})(s) &= k^{\tau-1}\sum_{n=1}^{\infty}\frac{n^{-s}}{n^{\tau}} - k^{-s}\sum_{n=1}^{\infty}\frac{n^{-s}}{n^{\tau}} \\ &= (k^{\tau-1}-k^{-s})\zeta(s+\tau) \\ &= k^{\tau-1}f_{k,\tau}(s) \quad \text{for } \mathrm{Re}(s) > 1/2,\end{aligned}$$

where $f_{k,\tau}(s) = (1-k^{1-s-\tau})\zeta(s+\tau)$.

3.1 A bounded operator from $\mathcal{H}^2$ to $H^2(\mathbb{C}_\alpha)$

The operator $\mathcal{M}_{1/s} : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2})$, defined by $\mathcal{M}_{1/s} : f(s) \mapsto f(s)/s$, is bounded according to [13]. For $\tau \geq 1/2$, we extend this construction by defining $\mathcal{M}^\tau : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2+\tau})$ via $\mathcal{M}^\tau : f \longmapsto f(s-\tau)/s$. To see that $\mathcal{M}^\tau$ is well defined and bounded, note that for $\sigma > \tau + 1/2$,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{|f(\sigma-\tau+it)|^2}{|\sigma+it|^2}\,dt \leq \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{|f(\sigma-\tau+it)|^2}{|\sigma-\tau+it|^2}\,dt \leq \|\mathcal{M}_{1/s}f\|_{1/2}^2,$$

and taking the supremum over $\sigma > \tau + 1/2$ yields

$$\|\mathcal{M}^\tau f\|_{1/2+\tau} \leq \|\mathcal{M}_{1/s}\|\,\|f\|_{\mathcal{H}^2}, \qquad \forall f \in \mathcal{H}^2.$$

The relevance of $\mathcal{M}^\tau$ to our approach lies in how it acts on the functions 1 and $f_{k,\tau}$. A direct computation gives

$$(\mathcal{M}^\tau 1)(s) = \frac{1}{s} = E(s),$$

and

$$(\mathcal{M}^\tau f_{k,\tau})(s) = \frac{f_{k,\tau}(s-\tau)}{s} = (1-k^{1-s})\frac{\zeta(s)}{s} = -kG_k(s).$$

3.2 Zero free region of ζ in $\mathbb{C}_\alpha$

We now establish one of the main results of the paper. The key step is the following generalization of Lemma 2.1 to the family $\{g_{k,\tau}\}$.

Lemma 3.1. *The series $\sum_{k=2}^{\infty}\frac{\mu(k)}{k^\tau}g_{k,\tau}$ converges to $-z$ in H_0^2, for each $\tau > 1/2$.*

Proof. Since $g_{1,\tau} \equiv 0$, the $k=1$ term may be included, giving

$$\begin{aligned}\sum_{k=2}^{n}\frac{\mu(k)}{k^\tau}g_{k,\tau}(z) &= \sum_{k=1}^{n}\frac{\mu(k)}{k^\tau}\left\{k^{\tau-1}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau}\right\} \\ &= \sum_{k=1}^{n}\frac{\mu(k)}{k}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{k=1}^{n}\frac{\mu(k)}{k^\tau}\sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau}.\end{aligned}$$

Since $\sum_{k=1}^{\infty} \frac{\mu(k)}{k} = 0$, the first term vanishes in the limit:

$$\left\| \sum_{k=1}^{n} \frac{\mu(k)}{k} \sum_{j=1}^{\infty} \frac{z^j}{j^\tau} \right\|_{H_0^2} = \left| \sum_{k=1}^{n} \frac{\mu(k)}{k} \right| \left\| \sum_{j=1}^{\infty} \frac{z^j}{j^\tau} \right\|_{H_0^2} \longrightarrow 0.$$

It therefore suffices to show that

$$\left\| -\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} + z \right\|_{H_0^2} \longrightarrow 0. \tag{3.1}$$

A rearrangement of the double sum gives

$$\begin{aligned} \sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} &= \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) \\ &= \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left[\frac{1}{j} \right] + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right), \end{aligned} \tag{3.2}$$

where the last equality uses the identity $\sum_{d|j} \mu(d) = \left[\frac{1}{j} \right]$. From (3.2),

$$\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} = z + \phi_n(z),$$

where

$$\phi_n(z) = \sum_{j=n+1}^{\infty} \left(\frac{1}{j^\tau} \sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right) z^j.$$

It remains to show that $\|\phi_n\|_{H_0^2} \to 0$ as $n \to \infty$. Letting $d(n)$ denote the number of divisors of n, we estimate

$$\|\phi_n\|_{H_0^2}^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left| \sum_{\substack{d|j, \\ 1 \le d \le n}} \mu(d) \right|^2 \le \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left(\sum_{d|j} 1 \right)^2 = \sum_{j=n+1}^{\infty} \frac{d(j)^2}{j^{2\tau}}.$$

Since $d(n) = o(n^\epsilon)$ for every $\epsilon > 0$, choosing $\epsilon > 0$ such that $\tau > 1/2 + \epsilon$ yields

$$\|\phi_n\|_{H_0^2}^2 \lesssim \sum_{j=n+1}^{\infty} \frac{j^{2\epsilon}}{j^{2\tau}} = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau - 2\epsilon}} \longrightarrow 0,$$

since $2\tau - 2\epsilon > 1$. □

For $n \geq 2$ write $S_n := \sum_{k=2}^{n} \mu(k) G_k$ for the n-th Möbius partial sum. With this notation, Lemma 3.1 transfers to the half-plane setting as follows.

Theorem 3.1. *The sequence (S_n) converges to E in $H^2(\mathbb{C}_{1/2+\tau})$, for every $\tau > 1/2$.*

Proof. By Lemma 3.1 and the isomorphism ψ,

$$\sum_{k=2}^{n} \frac{\mu(k)}{k} f_{k,\tau} \longrightarrow -1 \quad \text{in } \mathcal{H}^2.$$

Applying $\mathcal{M}^\tau$ and using $(\mathcal{M}^\tau f_{k,\tau})(s) = -kG_k(s)$, we conclude

$$S_n = \sum_{k=2}^{n} \mu(k) G_k \longrightarrow E \quad \text{in } H^2(\mathbb{C}_{1/2+\tau}). \tag{3.3}$$

□

Corollary 3.1. *E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$.*

The membership of E in the closed linear span of $\{G_k : k \geq 2\}$ also has direct consequences for the location of the zeros of ζ.

Proposition 3.1. *For any $\alpha \geq 1/2$,*

$$E \in \overline{\text{span}}\{G_k : k \geq 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \neq 0, \quad \forall s \in \mathbb{C}_\alpha.$$

Proof. Let ρ be a zero of ζ in $\mathbb{C}_\alpha$. Then $G_k(\rho) = 0$ for every $k \geq 2$. Since $H^2(\mathbb{C}_\alpha)$ is a reproducing kernel Hilbert space, convergence in norm implies pointwise convergence, so $E(\rho) = 1/\rho = 0$, which is impossible. □

3.3 A lower bound from the zeros of ζ

Proposition 3.1 is qualitative: it turns convergence into a zero-free half-plane. The reproducing kernel of $H^2(\mathbb{C}_\alpha)$ yields a quantitative companion in the opposite direction, in which a single zero of ζ inside $\mathbb{C}_\alpha$ produces an explicit, n-independent floor for the approximation error. For $n \geq 2$ and $\alpha > 0$, define the *optimal approximation error*

$$d_n(\alpha)^2 := \inf_{c_2,\dots,c_n \in \mathbb{C}} \Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2,$$

the best approximation of E by the first $n-1$ generators. The restriction $\alpha > 0$ is precisely what is needed for E and the G_k to belong to $H^2(\mathbb{C}_\alpha)$: both are analytic on $\mathbb{C}_\alpha$, the factor $k^{-s} - k^{-1}$ vanishing at $s = 1$ and so cancelling the pole of ζ, whereas for $\alpha \leq 0$ they have a pole at $s = 0$. Taking $c_k = \mu(k)$ shows $d_n(\alpha)^2 \leq \|S_n - E\|_\alpha^2$, so any lower bound for $d_n(\alpha)^2$ is a fortiori one for the Möbius error.

Theorem 3.2. *Let $\alpha > 0$ and suppose ζ has a zero ρ with $\mathrm{Re}(\rho) > \alpha$. Then for every $n \geq 2$ and all coefficients $c_2, \ldots, c_n \in \mathbb{C}$,*

$$\Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2 \;\geq\; \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2} \;=\; \frac{2\,(\mathrm{Re}\,\rho - \alpha)}{|\rho|^2} \;>\; 0.$$

More generally, if $\rho_1, \ldots, \rho_m$ are zeros of ζ in $\mathbb{C}_\alpha$, then for every $n \geq 2$,

$$d_n(\alpha)^2 \;\geq\; v^* K^{-1} v, \qquad K = \Big(\tfrac{1}{\rho_i + \overline{\rho_j} - 2\alpha}\Big)_{i,j=1}^{m}, \qquad v = \big(1/\rho_i\big)_{i=1}^{m}.$$

In particular $d_n(\alpha)^2$ is bounded below by a positive constant independent of n, so $d_n(\alpha)^2 \not\to 0$.

Proof. Since $\mathrm{Re}(\rho) > \alpha$ we have $\rho \in \mathbb{C}_\alpha$, and $\zeta(\rho) = 0$ gives $G_k(\rho) = (k^{-\rho} - k^{-1})\zeta(\rho)/\rho = 0$ for every k. Hence $F := E - \sum_{k=2}^{n} c_k G_k$ satisfies $F(\rho) = E(\rho) = 1/\rho$. By the reproducing property $F(\rho) = \langle F, k_\rho \rangle_\alpha$, so the Cauchy–Schwarz inequality gives

$$\|F\|_\alpha^2 \;\geq\; \frac{|\langle F, k_\rho \rangle_\alpha|^2}{\|k_\rho\|_\alpha^2} = \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2}.$$

With $\|k_\rho\|_\alpha^2 = k_\rho(\rho) = (2\,\mathrm{Re}\,\rho - 2\alpha)^{-1}$ and $|E(\rho)|^2 = |\rho|^{-2}$, this is the first inequality.

For the general case, let $N = \{g \in H^2(\mathbb{C}_\alpha) : g(\rho_i) = 0,\ 1 \leq i \leq m\}$, whose orthogonal complement is $V = \mathrm{span}\{k_{\rho_1}, \ldots, k_{\rho_m}\}$. Every combination $\sum_k c_k G_k$ lies in N, since $G_k(\rho_i) = 0$; thus, with P the orthogonal projection onto V,

$$\Big\| E - \sum_{k} c_k G_k \Big\|_\alpha \;\geq\; \Big\| P\big(E - \textstyle\sum_k c_k G_k\big) \Big\|_\alpha = \|PE\|_\alpha.$$

A standard computation with the kernels gives $\|PE\|_\alpha^2 = v^* K^{-1} v$, where $K_{ij} = \langle k_{\rho_j}, k_{\rho_i} \rangle_\alpha = k_{\rho_j}(\rho_i) = (\rho_i + \overline{\rho_j} - 2\alpha)^{-1}$ and $v_i = \langle E, k_{\rho_i} \rangle_\alpha = \overline{E(\rho_i)}$. Taking the infimum over c yields the bound on $d_n(\alpha)^2$; the right-hand side is a fixed positive constant, so $d_n(\alpha)^2 \not\to 0$. □

Corollary 3.2. *Let $\Theta = \sup\{\mathrm{Re}\,\rho : \zeta(\rho) = 0\}$ and $\alpha_c = \inf\big\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\big\}$. Then $\tfrac{1}{2} \leq \Theta \leq \alpha_c \leq 1$, and $\alpha_c = \tfrac{1}{2}$ if and only if RH holds.*

Proof. By Theorem 3.1 the set $\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\}$ is nonempty and bounded below by $\tfrac{1}{2}$, so α_c is well-defined and $\alpha_c \leq 1$. For $\alpha < \Theta$ pick a zero ρ with $\mathrm{Re}\,\rho > \alpha$; Theorem 3.2 gives $d_n(\alpha)^2 \not\to 0$, whence $\Theta \leq \alpha_c$. As $\Theta \geq \tfrac{1}{2}$ is classical, $\tfrac{1}{2} \leq \Theta \leq \alpha_c \leq 1$. If $\alpha_c = \tfrac{1}{2}$ then $\Theta = \tfrac{1}{2}$, which is RH. Conversely, under RH, Theorem 2.2 places E in the closed linear span of $\{G_k\}_{k \geq 2}$ in $H^2(\mathbb{C}_{1/2})$. Since $d_n(1/2)$ is non-increasing in n, this forces $d_n(1/2)^2 \to 0$; hence $\alpha = \tfrac{1}{2}$ belongs to the set $\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\}$, so that $\alpha_c \leq \tfrac{1}{2}$. Combined with $\Theta \geq \tfrac{1}{2}$, this gives $\alpha_c = \tfrac{1}{2}$. □

4 Numerical experiments

The range $1/2 \leq \alpha \leq 1$ left open by Theorem 3.1 lies beyond the reach of our analytical methods; this section probes it numerically. Two questions guide the experiments: how the Möbius error $s_n(\alpha) = \|S_n - E\|_\alpha^2$ behaves as $n \to \infty$ across the strip, and whether the obstruction on the critical line traced by Corollary 3.2 is intrinsic to the subspaces $\mathrm{span}\{G_2, \ldots, G_n\}$ or only to the fixed choice of coefficients $\mu(k)$. The first question is addressed in Subsection 4.2 and the second in Subsection 4.3. Both rest on the computation of norms in $H^2(\mathbb{C}_\alpha)$, which we describe first.

4.1 Computation of the norms

A direct evaluation of the boundary integral

$$s_n(\alpha) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \bigl|S_n(\alpha+it) - E(\alpha+it)\bigr|^2\,dt$$

by numerical quadrature is costly and unreliable for large n: every evaluation of the integrand involves $\zeta(\alpha+it)$ and the $n-1$ oscillating factors $k^{-\alpha-it}$, and the integrand oscillates on scales that shrink as n and $|t|$ grow. We avoid quadrature altogether by means of the following exact representation of $s_n(\alpha)$ as a series with elementary terms.

Proposition 4.1. *Let $\alpha > 0$ and $n \geq 2$. For $m \geq 1$ put*

$$R_n(m) = 1 + \sum_{k=2}^{n} \mu(k)\,\frac{m \bmod k}{k} \qquad \text{and} \qquad w_m(\alpha) = \int_m^{m+1} \frac{du}{u^{2\alpha+1}} = \frac{m^{-2\alpha} - (m+1)^{-2\alpha}}{2\alpha}.$$

Then

$$s_n(\alpha) = \sum_{m=1}^{\infty} w_m(\alpha)\,R_n(m)^2. \tag{4.1}$$

Proof. Write $\{y\}$ for the fractional part of y and, for $k \geq 2$, let $\rho_k(x) = \{1/(kx)\} - \{1/x\}/k$ for $x > 0$. If $x > 1$, both $1/(kx)$ and $1/x$ lie in $(0,1)$, so $\rho_k(x) = 0$; moreover $|\rho_k| \leq 2$. For $0 < \operatorname{Re}(s) < 1$ we have the classical formula [15, §2.1], written here after the change of variables $x \mapsto 1/x$,

$$\int_0^{\infty} \{1/x\}\,x^{s-1}\,dx = -\frac{\zeta(s)}{s},$$

and the substitution $y = kx$ gives $\int_0^{\infty}\{1/(kx)\}\,x^{s-1}\,dx = -k^{-s}\zeta(s)/s$. Subtracting the two identities,

$$\int_0^1 \rho_k(x)\,x^{s-1}\,dx = -\bigl(k^{-s} - k^{-1}\bigr)\frac{\zeta(s)}{s} = -G_k(s), \qquad 0 < \operatorname{Re}(s) < 1.$$

Since ρ_k is bounded on $(0,1)$, the left-hand side converges absolutely and is analytic on $\operatorname{Re}(s) > 0$, so by analytic continuation the identity holds on the whole half-plane $\mathbb{C}_0$. Together with $E(s) = \int_0^1 x^{s-1}\,dx$ this gives

$$E(s) - S_n(s) = \int_0^1 f_n(x)\,x^{s-1}\,dx, \qquad f_n = 1 + \sum_{k=2}^{n}\mu(k)\,\rho_k, \qquad \operatorname{Re}(s) > 0.$$

Let $\sigma \geq \alpha$. On the line $\operatorname{Re}(s) = \sigma$ the substitution $x = e^{-y}$ turns this integral into the Fourier transform of $g_\sigma(y) = f_n(e^{-y})\,e^{-\sigma y}$, $y > 0$, which lies in $L^1(0,\infty)\cap L^2(0,\infty)$ because f_n is bounded and $\sigma > 0$. By Plancherel's theorem [14, Theorem 9.13],

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\bigl|(E - S_n)(\sigma+it)\bigr|^2\,dt = \int_0^{\infty}|g_\sigma(y)|^2\,dy = \int_0^1 |f_n(x)|^2\,x^{2\sigma-1}\,dx.$$

Since $0 < x < 1$, the right-hand side is a non-increasing function of σ, and it tends to its value at $\sigma = \alpha$ as $\sigma \to \alpha^+$ by monotone convergence. Hence $E - S_n \in H^2(\mathbb{C}_\alpha)$ and, by the definition of the norm,

$$s_n(\alpha) = \sup_{\sigma>\alpha} \int_0^1 |f_n(x)|^2 \, x^{2\sigma-1}\, dx = \int_0^1 |f_n(x)|^2 \, x^{2\alpha-1}\, dx.$$

Finally put $x = 1/u$, so that the last integral equals $\int_1^\infty |f_n(1/u)|^2 \, u^{-2\alpha-1}\, du$. For $u \in [m, m+1)$ the interval $(m, u]$ contains no integer, hence no multiple of k; thus $\{u/k\} = u/k - \lfloor m/k \rfloor$ and $\{u\} = u - m$, which give $\rho_k(1/u) = m/k - \lfloor m/k \rfloor = (m \bmod k)/k$ and $f_n(1/u) = R_n(m)$. Splitting the integral over the intervals $[m, m+1)$ yields (4.1). □

Remark 4.1. *(i) For $\alpha = 1/2$ the weights are $w_m(1/2) = 1/(m(m+1))$, and Proposition 4.1 turns the approximation of E in $H^2(\mathbb{C}_{1/2})$ into an approximation problem for the constant sequence $\mathbf{1}$ by the sequences r_k/k, $r_k(m) = m \bmod k$, in a weighted ℓ^2-space; compare with the setting of Balazard and Saias [4] recalled in the introduction. The proof also shows that $s_n(1/2) = \|f_n\|^2_{L^2(0,1)}$, which makes explicit the link with the $L^2(0,1)$ setting of Báez-Duarte.*

(ii) Since $\sum_{k\le m} \mu(k)\lfloor m/k \rfloor = 1$ for every $m \ge 1$ [1, Theorem 3.12], and $\lfloor m/k \rfloor = 0$ for $k > m$, for $m \le n$ we get

$$R_n(m) = 1 + \sum_{k=1}^{n} \mu(k)\Big(\frac{m}{k} - \Big\lfloor \frac{m}{k} \Big\rfloor\Big) = m\, M_1(n), \qquad M_1(n) = \sum_{k\le n} \frac{\mu(k)}{k},$$

and therefore

$$s_n(\alpha) \ge M_1(n)^2 \sum_{m=1}^{n} m^2\, w_m(\alpha), \qquad \sum_{m=1}^{n} m^2\, w_m(\alpha) \sim \frac{n^{2-2\alpha}}{2-2\alpha} \quad \Big(\tfrac{1}{2} \le \alpha < 1\Big). \tag{4.2}$$

Thus the decay of $s_n(\alpha)$ is limited from below by the size of the partial sums $M_1(n)$ of $\sum_k \mu(k)/k$. Under RH one has $M_1(n) = O(n^{-1/2+\varepsilon})$ (cf. [15, Ch. XIV]); if $M_1(n)$ is of exact order $n^{-1/2}$ along a sequence of values of n, the right-hand side of (4.2) is of order $n^{1-2\alpha}$ along that sequence.

Implementation. We evaluate the series (4.1) in three parts. The terms with $m \le m_0$ are summed directly. On $m_0 < m \le M$ the terms are grouped into blocks $[a, b)$ of relative length $(b-a)/a \le h$, and the weight is expanded to second order about the centre c of each block,

$$\sum_{a\le m<b} w_m(\alpha)\, R_n(m)^2 \approx \sum_{j=0}^{2} \frac{w^{(j)}(c)}{j!} \sum_{a\le m<b} (m-c)^j R_n(m)^2, \qquad w(x) = \frac{x^{-2\alpha} - (x+1)^{-2\alpha}}{2\alpha},$$

which has relative error $O(h^3)$ and reduces the dependence on α to three moments of R_n^2 per block, shared by all values of α. For $m > M$ we replace $R_n(m)^2$ by its mean value $\overline{R_n^2}$ over a period of $m \mapsto R_n(m)$, so that the tail contributes $\overline{R_n^2}\,(M+1)^{-2\alpha}/(2\alpha)$. This mean is known exactly: for m uniformly distributed modulo $\operatorname{lcm}(j,k)$, the Chinese remainder theorem shows that the residues $m \bmod j$ and $m \bmod k$ are correlated only through $m \bmod \gcd(j,k)$, with covariance

$(\gcd(j,k)^2-1)/12$, and writing $\gcd(j,k)^2=\sum_{d\mid\gcd(j,k)}J_2(d)$, where J_2 is Jordan's totient, one finds

$$\overline{R_n^2}=\Big(1+\sum_{k=2}^{n}\mu(k)\,\frac{k-1}{2k}\Big)^2+\frac{1}{12}\sum_{d\ge 2}J_2(d)\Big(\sum_{\substack{2\le k\le n\\ d\mid k}}\frac{\mu(k)}{k}\Big)^2.$$

Since $R_n(m)=R_{n-1}(m)+\mu(n)\,(m \bmod n)/n$, passing from $n-1$ to n costs a single pass over $m\le M$, and only the n with $\mu(n)\neq 0$ contribute. All the values $s_n(\alpha)$ with $n\le N$ are therefore obtained with $O(NM)$ elementary operations, without evaluating ζ and without numerical quadrature. The computation was organised so that each block of values of m stays in cache while it runs through all n, the blocks being processed in parallel. A Fortran 2008 program with OpenMP computed $s_n(\alpha)$ for the 51 values $\alpha=0.50,0.51,\dots,1.00$ and 3001 logarithmically spaced values of $n\in[10^3,10^6]$ in 40 minutes on an Intel Core i7-13700KF processor (24 threads), with $m_0=10^4$, $h=10^{-3}$ and $M=3\cdot 10^8$. The block expansion changes the results by less than 10^{-9} (relative) with respect to direct summation. Repeating the computation with $M=10^8$ changes them by at most $3.5\cdot 10^{-4}$, the largest differences occurring for n close to 10^6, and runs with $M=10^7,10^8,10^9$ for $n\le 10^5$ show that the truncation error decreases roughly like $M^{-1.8}$; we therefore estimate the relative error of the values reported below to be smaller than 10^{-4}. As independent checks, the representation (4.1) reproduces to ten digits the identity $\langle E,G_k\rangle_{1/2}=G_k(1)=-(\log k)/k$, which holds because E is the reproducing kernel of $H^2(\mathbb{C}_{1/2})$ at $s=1$, and it agrees with direct quadrature of the boundary integral within the accuracy of the latter, which is typically of one percent.

Optimal error. The optimal error of Subsection 3.3 is computed on the critical line as $d_n(1/2)^2=\|E\|_{1/2}^2-b^{\top}A^{-1}b$, with Gram data $A=(\langle G_j,G_k\rangle_{1/2})_{j,k=2}^{n}$ and $b=(\langle E,G_k\rangle_{1/2})_{k=2}^{n}$. Because E and the G_k are real on the real axis, they satisfy the reflection symmetry $G_k(\bar s)=\overline{G_k(s)}$ (and likewise for E), so each such inner product reduces to a real half-line integral,

$$\langle G_j,G_k\rangle_{1/2}=\frac{1}{\pi}\int_0^{\infty}\operatorname{Re}\big[G_j(\tfrac12+it)\,\overline{G_k(\tfrac12+it)}\big]\,dt,$$

and likewise for $\langle E,G_k\rangle_{1/2}$; hence A and b are real and the minimising coefficients may be taken real. These boundary integrals were evaluated on a fixed quadrature grid of approximately 2.8×10^4 nodes, refined near $t=0$ and with a geometric tail out to $t=3\times 10^5$. On this grid the exact value $\|E\|_{1/2}^2=1/(2\alpha)=1$ is reproduced to six digits, and the computed values of d_n^2 are unchanged, to plotting accuracy, under refinement of the grid; since the Gram matrix grows ill-conditioned with n (its condition number is the square of that of the underlying design matrix), the linear systems were solved only in the numerically stable range $n\le 5120$. These computations were carried out in Python 3.13, with the Riemann zeta function taken from `scipy.special.zeta`, on a machine with a 5.2 GHz Intel Core i9-12900K processor and 128 GB of DDR4 RAM. The code generating all figures and tables of this section is available from the authors upon request.

4.2 The Möbius error across the strip

We computed $s_n(\alpha)$ for $\alpha = 0.50, 0.51, \ldots, 1.00$ and $10^3 \le n \le 10^6$, at 1000 logarithmically spaced values of n per decade. Figure 1 shows the sequences for $\alpha = 0.50, 0.60, \ldots, 1.00$. They oscillate strongly with n: within each decade the largest value exceeds the smallest by a factor of about 8 on the critical line, and by a factor of several hundred for $\alpha = 1$. The oscillations occur at the same values of n for all α, and the value of $s_n(\alpha)$ at an isolated n is therefore not representative of its behaviour. We summarise each sequence instead by its largest and smallest values over each decade of n (Table 1), and measure its decay by the change of $\log_{10} \max s_n$ and $\log_{10} \min s_n$ per decade between the first and the last decade (Figure 2).

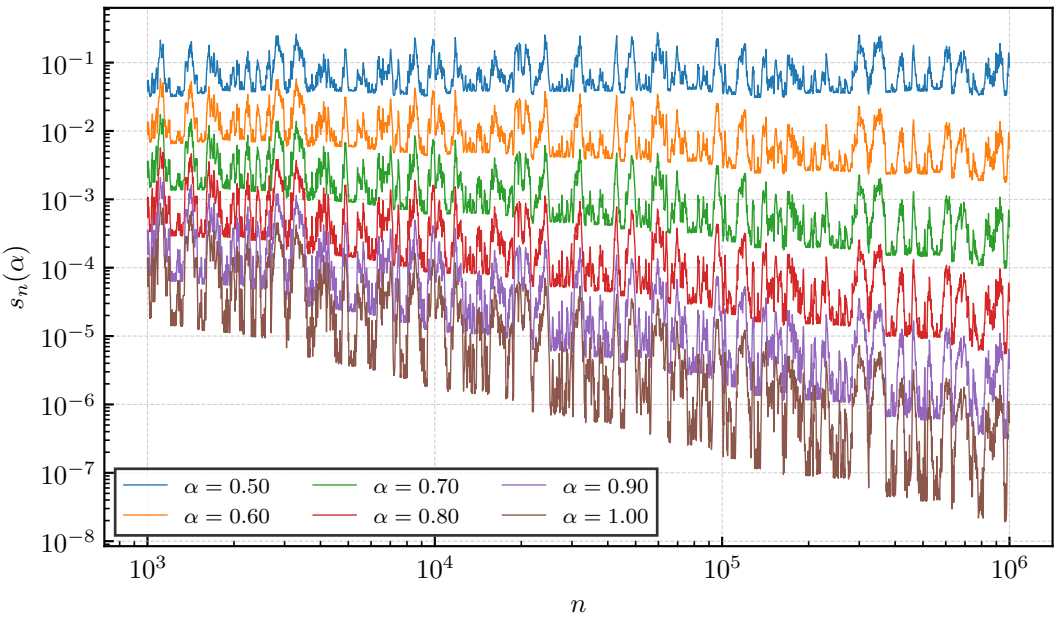


Figure 1: The Möbius error $s_n(\alpha)$ for $\alpha = 0.50, 0.60, \ldots, 1.00$ and $10^3 \le n \le 10^6$, in log–log scale.

On the critical line the Möbius error does not decay: over the three decades the maxima of $s_n(1/2)$ are 0.262, 0.277 and 0.253, and the minima $3.18 \cdot 10^{-2}$, $3.41 \cdot 10^{-2}$ and $3.08 \cdot 10^{-2}$. This is the behaviour forced by the result of Báez-Duarte [3], by which $s_n(1/2) \not\to 0$. For every $\alpha > 1/2$, on the other hand, both the maxima and the minima decrease geometrically from one decade to the next, and Figure 2 shows that the rate is very close to $10^{1-2\alpha}$ per decade: over the whole range the observed changes of $\log_{10}$ per decade differ from $1 - 2\alpha$ by less than 0.08. The data are thus consistent with

$$s_n(\alpha) \asymp n^{1-2\alpha}, \qquad \tfrac{1}{2} \le \alpha \le 1,$$

for the envelope of the oscillations, and in particular with $s_n(\alpha) \to 0$ for every $\alpha > 1/2$. The convergence is very slow near the critical line: for $\alpha = 0.55$ the error decreases only by a factor $10^{0.1} \approx 1.26$ per decade, so that for moderate n the behaviour for α slightly above $1/2$ is hard to tell apart from that on the critical line. The exponent $1 - 2\alpha$ is precisely the one given by

α	$10^3 \le n < 10^4$		$10^4 \le n < 10^5$		$10^5 \le n \le 10^6$	
	$\max s_n$	$\min s_n$	$\max s_n$	$\min s_n$	$\max s_n$	$\min s_n$
0.50	2.617e−1	3.181e−2	2.772e−1	3.408e−2	2.530e−1	3.079e−2
0.55	1.219e−1	1.189e−2	9.722e−2	1.086e−2	7.554e−2	7.605e−3
0.60	5.866e−2	4.390e−3	3.954e−2	3.213e−3	2.298e−2	1.776e−3
0.65	3.162e−2	1.632e−3	1.683e−2	9.559e−4	7.140e−3	4.171e−4
0.70	1.740e−2	6.106e−4	7.338e−3	2.857e−4	2.393e−3	9.806e−5
0.75	9.798e−3	2.296e−4	3.291e−3	8.578e−5	8.578e−4	2.316e−5
0.80	5.662e−3	8.677e−5	1.528e−3	2.584e−5	3.197e−4	5.499e−6
0.85	3.370e−3	3.293e−5	7.396e−4	7.811e−6	1.254e−4	1.313e−6
0.90	2.073e−3	1.255e−5	3.768e−4	2.368e−6	5.263e−5	3.157e−7
0.95	1.323e−3	4.793e−6	2.039e−4	7.196e−7	2.405e−5	7.693e−8
1.00	8.784e−4	1.838e−6	1.181e−4	2.192e−7	1.217e−5	1.926e−8

Table 1: Largest and smallest values of the Möbius error $s_n(\alpha)$ over each decade of n, for $\alpha = 0.50, 0.55, \ldots, 1.00$, computed at 3001 logarithmically spaced values of $n \in [10^3, 10^6]$.

the lower bound (4.2) when $M_1(n)$ is of order $n^{-1/2}$, and the largest peaks of $s_n(\alpha)$ in Figure 1 occur where $nM_1(n)^2$ is large.

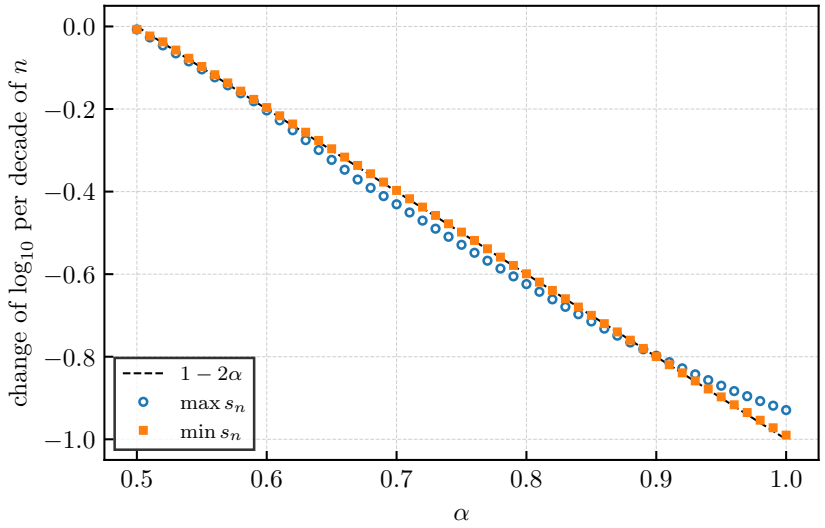


Figure 2: Change of $\log_{10} \max s_n(\alpha)$ and of $\log_{10} \min s_n(\alpha)$ per decade of n, from $10^3 \le n < 10^4$ to $10^5 \le n \le 10^6$, for $\alpha = 0.50, 0.51, \ldots, 1.00$, compared with $1 - 2\alpha$ (dashed).

These observations must be read with care. By Proposition 3.1, a proof that $s_n(\alpha) \to 0$ for some $\alpha < 1$ would give a zero-free half-plane $\mathbb{C}_\alpha$, far beyond the known zero-free regions of ζ. The behaviour observed is the one expected under RH, under which $M_1(n) = O(n^{-1/2+\varepsilon})$, but no finite computation can rule out the alternative: by Theorem 3.2, a zero ρ with $\operatorname{Re}\rho > \alpha$ only forces $s_n(\alpha) \ge 2(\operatorname{Re}\rho - \alpha)/|\rho|^2$, and since every zero with $|\operatorname{Im}\rho| \le 3 \cdot 10^{12}$ lies on the critical line [12], this floor would be smaller than 10^{-24}.

4.3 The optimal error on the critical line

The critical line is the one place in the range studied where S_n demonstrably fails: by Báez-Duarte [3] the Möbius coefficients do not sum to E at $\alpha = 1/2$, and the undamped oscillations of Figure 1 are the price of this fixed choice. Since it is exactly at $\alpha = 1/2$ that Theorem 2.2 formulates the RH, the natural question is what this failure rules out. The answer is: nothing about the subspaces $\operatorname{span}\{G_2, \ldots, G_n\}$ themselves, since $\mu(k)$ is only one admissible choice of coefficients, and it is the subspaces, not the choice, that Theorem 3.2 ties to the zeros of ζ. This is what motivates the present experiment: we replace $s_n(1/2)$ by the optimal error $d_n(1/2)^2$ of Subsection 3.3, the true distance from E to $\operatorname{span}\{G_2, \ldots, G_n\}$ on the critical line, which removes the coefficients from the picture by optimising over all of them. It is computed by the Gram matrix method of Subsection 4.1.

Figure 3 contrasts the two sequences, and Table 2 records their values at selected n. While the Möbius error $s_n(1/2)$ remains near $10^{-1.5}$ with persistent oscillations, the optimal error d_n^2 decreases steadily, though very slowly, and satisfies $d_n^2 \leq s_n(1/2)$ throughout the computed range. The "divergence" on the critical line is therefore a statement about the coefficients rather than about the subspaces: the obstruction lies in the fixed choice $c_k = \mu(k)$, while the true distance from E to $\operatorname{span}\{G_2, \ldots, G_n\}$ continues to decrease over the range accessible to stable computation.

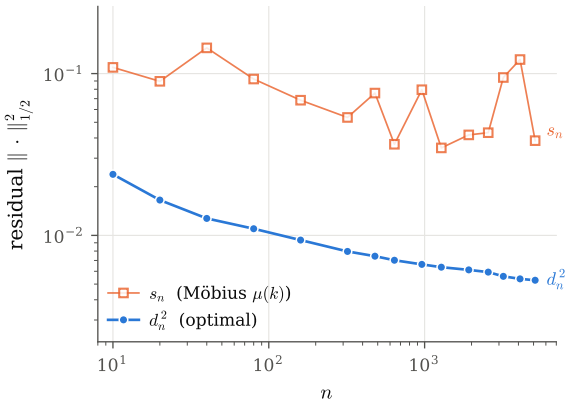


Figure 3: The optimal error $d_n(1/2)^2$ (blue) and the Möbius error $s_n(1/2)$ (red) on the critical line, $2 \leq n \leq 5120$, in log–log scale.

n	$d_n(1/2)^2$	$s_n(1/2)$
40	1.273e−2	1.442e−1
160	9.355e−3	6.846e−2
640	7.022e−3	3.657e−2
1280	6.361e−3	3.473e−2
2560	5.931e−3	4.312e−2
5120	5.282e−3	3.850e−2

Table 2: The same two errors at selected n; in every case $d_n^2 \leq s_n(1/2)$.

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